3.11 Sorting (Lab 3.3)

LAB ACTIVITY | 3.11.1: Sorting (Lab 3.3) | Full screen | 0 / 30

First, download this template and rename it `sorting.py`:

- template.py

Write code that allows the user to enter four numbers and stores these numbers in a list. The program will then create and print a new list with the same numbers in sorted order (from min to max).

Please note the following:

- Prompt the user for each number in turn using `number:`.
- The only functions on lists and list methods you may use are:
  - `lst.append(x)`
  - `min(lst)`
  - `max(lst)`
  - `lst.remove(item)`
  - `lst.pop(index)`
  - `len(lst)`
  - `list1 + list2`

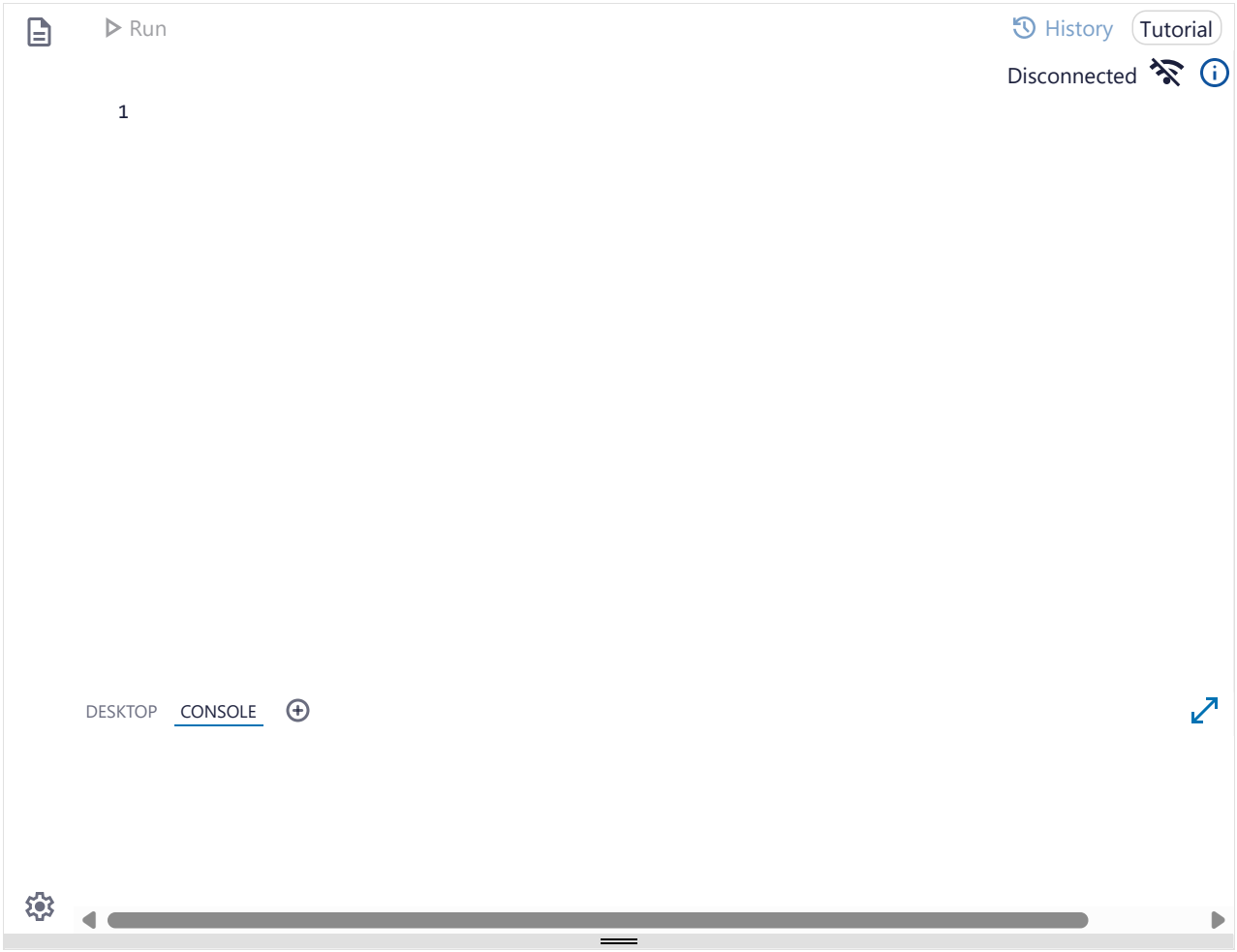
  If any of these operations are unfamiliar to you, we suggest that you try them out!
- We suggest that you use the "selection sort"; algorithm, which takes the given list of unsorted values, creates a second list for sorted values (initially empty) and repeatedly moves the *smallest* item from the unsorted list to the sorted list.
- End by printing the sorted list.
- If you're have problems creating the algorithm, consider what you'd do if the list had only one number. Two numbers? Three numbers?
- Normally you'd use `lst.sort()` or `sorted(lst)` for this task, but **for this exercise, do not use them**.

Begin by writing an ***algorithm*** for your selection sort as a comment in your file. Then implement your algorithm using the given list of valid Python commands.

Here is an example user interaction:

```
number: 0.0↵
number: 1.0↵
number: 5.0↵
number: 3.0↵
[0.0, 1.0, 3.0, 5.0]
```

Open new tab | Dock



Submit for grading

Coding trail of your work What is this?

```
History of your effort will appear here once you begin working on this
zyLab.
```

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Template, Model Solution, and Test Bench Show

Lab statistics Show

Student behavior insights (beta) Show

Similarity detection Show

Student activity Show